Math 510 **Fall 2026**
Homework 1 **Drew Armstrong**

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1. Explicitly use the axioms of vector spaces to prove that

$$0\mathbf{x} = \mathbf{0} \text{ for all vectors } \mathbf{x},$$

where 0 is the zero scalar and $\mathbf{0}$ is the zero vector.

2. For any vectors $\mathbf{u}, \mathbf{v}$ and scalar t, explicitly use the axioms of inner products to show that

$$\langle \mathbf{u} - t\mathbf{v}, \mathbf{u} - t\mathbf{v} \rangle = \langle \mathbf{u}, \mathbf{u} \rangle - 2t\langle \mathbf{u}, \mathbf{v} \rangle + t^2 \langle \mathbf{v}, \mathbf{v} \rangle.$$

3. Let V be a vector space and consider a list of vectors $\mathbf{u}_1, \ldots, \mathbf{u}_k \in V$. Prove that the following two conditions are equivalent:

- For any $\mathbf{v} \in V$ there exists at most one list of scalars $a_1, \ldots, a_k$ satisfying
$$\mathbf{v} = a_1\mathbf{u}_1 + \cdots + a_k\mathbf{u}_k.$$
- For any list of scalars $a_1, \ldots, a_k$ we have
$$a_1\mathbf{u}_1 + \cdots + a_k\mathbf{u}_k = \mathbf{0} \quad \Longrightarrow \quad a_1 = \cdots = a_k = 0.$$

  In either case we say that the vectors $\mathbf{u}_1, \ldots, \mathbf{u}_k$ are *linearly independent.* Show that the zero vector cannot be a member of any linearly independent set.

4. Let V be an inner product space (over $\mathbb{R}$). By definition we have $\langle \mathbf{v}, \mathbf{v} \rangle \geq 0$, hence there exists a unique non-negative square root, which we denote by

$$\|\mathbf{v}\| := \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}.$$

For any scalar $a \in \mathbb{R}$, prove that $\|a\mathbf{v}\| = |a|\|\mathbf{v}\|$.

5. Let V be an inner product space (over $\mathbb{R}$). Prove the Cauchy-Schwarz inequality:

$$|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\|\|\mathbf{v}\|.$$

Hint: For any scalar $t \in \mathbb{R}$, show that

$$\langle \mathbf{u}, \mathbf{u} \rangle - 2t\langle \mathbf{u}, \mathbf{v} \rangle + t^2\langle \mathbf{v}, \mathbf{v} \rangle = \langle \mathbf{u} - t\mathbf{v}, \mathbf{u} - t\mathbf{v} \rangle \geq 0.$$

Substitute $t = \langle \mathbf{u}, \mathbf{v} \rangle / \langle \mathbf{v}, \mathbf{v} \rangle$ into the inequality (we assume $\mathbf{v} \neq \mathbf{0}$) and then multiply both sides of the inequality by $\langle \mathbf{v}, \mathbf{v} \rangle$.

Challenge Problem (Steinitz Exchange). Let $\mathbf{u}_1, \ldots, \mathbf{u}_k$ be an independent set and let $\mathbf{v}_1, \ldots, \mathbf{v}_\ell$ be a spanning set for a vector space V. Prove that $k \leq \ell$.

Hint: Since the v_i are spanning we can write $\mathbf{u}_1 = a_1\mathbf{v}_1 + \cdots + a_\ell\mathbf{v}_\ell$ for some a_i. Since $\mathbf{u}_1 \neq \mathbf{0}$ there exists some $a_j \neq 0$. After relabeling the $\mathbf{v}_i$ we may assume that a_1, so that

$$\mathbf{v}_1 = \frac{1}{a_1}\mathbf{u}_1 - \frac{a_2}{a_1}\mathbf{v}_2 - \cdots - \frac{a_\ell}{a_1}\mathbf{v}_\ell.$$

Prove that $\mathbf{u}_1, \mathbf{v}_2, \ldots, \mathbf{v}_\ell$ is still a spanning set, hence we can write

$$\mathbf{u}_2 = b_1\mathbf{u}_1 + b_2\mathbf{u}_2 + \cdots + b_\ell\mathbf{v}_\ell.$$

Since $\mathbf{u}_1$ and $\mathbf{u}_2$ are independent (any subset of an independent set is independent) there exists $b_j \neq 0$ with $j \geq 2$. After relabeling we may assume that $b_2 \neq 0$. Use this to show that $\mathbf{u}_1, \mathbf{u}_2, \mathbf{v}_3, \ldots, \mathbf{v}_\ell$ is still a spanning set. Continue.